Task 1: Project Setup and Introduction

```
•[1]:
# Name: ishika
# Roll No: 2301010303
# Course: image processing and computer vision
# Unit: unit 1
# Assignment Title: Smart Document Scanner & Quality Analysis System
```

```
[2]:
print("   Welcome! my name is ishika")
```

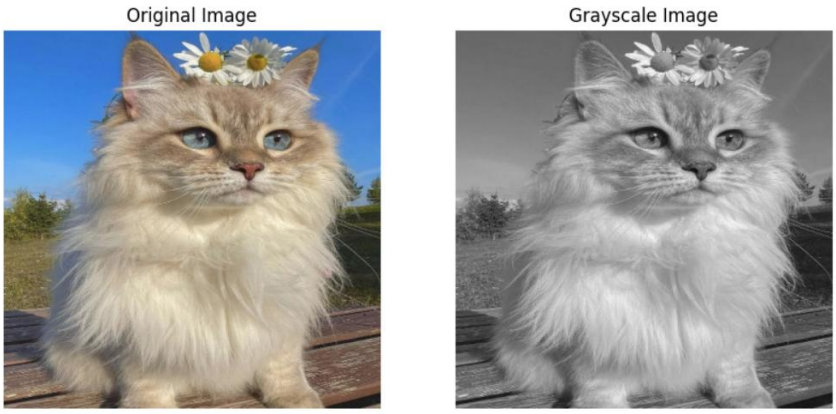
```
Welcome! my name is ishika
```

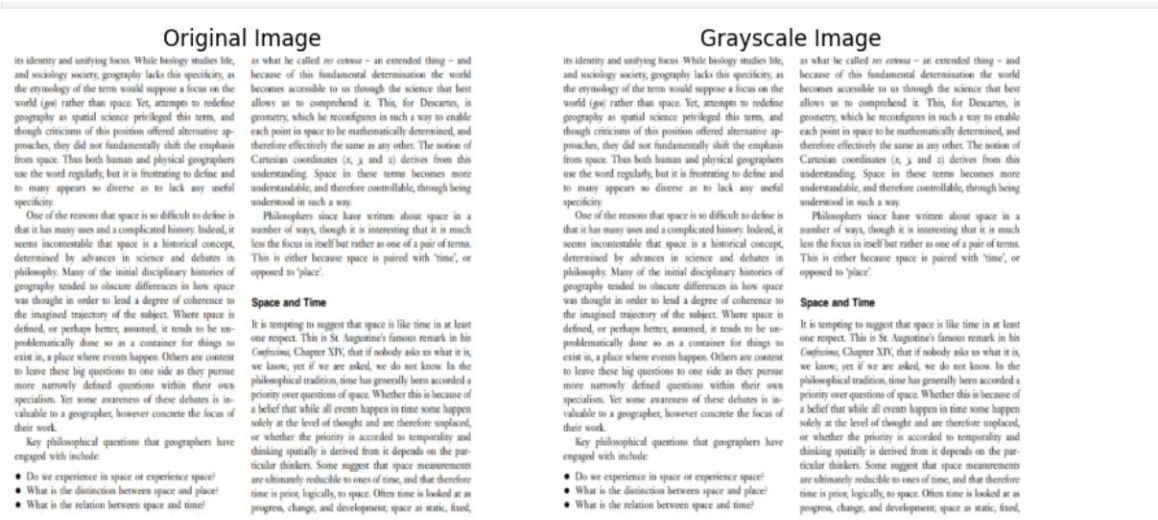
```
[ ]:
```

Task 2: Image Acquisition

```
import cv2
import matplotlib.pyplot as plt

# Load image
image = cv2.imread("doc.jpeg")

if image is None:
    print("Image not found! Check filename.")
else:
    # Resize image
    resized = cv2.resize(image, (512, 512))

    # Convert to grayscale
    gray = cv2.cvtColor(resized, cv2.COLOR_BGR2GRAY)
```

```
# Show images
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.title("Original Image")
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.axis("off")

plt.subplot(1,2,2)
plt.title("Grayscale Image")
plt.imshow(gray, cmap="gray")
plt.axis("off")

plt.show()
```



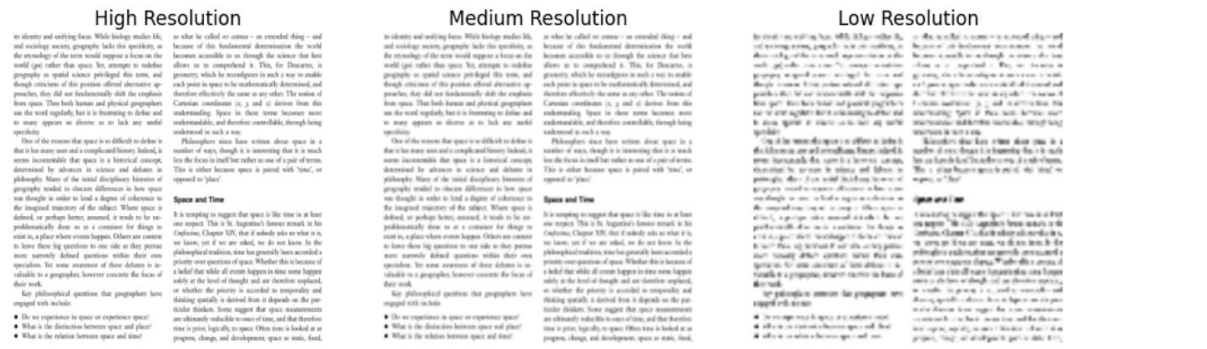


Task 3: Image Sampling (Resolution Analysis)

```
import cv2

import matplotlib.pyplot as plt


# Down-sampling

high = cv2.resize(gray, (512, 512))

medium = cv2.resize(gray, (256, 256))

low = cv2.resize(gray, (128, 128))


# Upscale back for viewing

medium_up = cv2.resize(medium, (512, 512))

low_up = cv2.resize(low, (512, 512))


# Display comparison

plt.figure(figsize=(12,6))
```

```
plt.subplot(1,3,1)
plt.title("High Resolution")
plt.imshow(high, cmap="gray")
plt.axis("off")

plt.subplot(1,3,2)
plt.title("Medium Resolution")
plt.imshow(medium_up, cmap="gray")
plt.axis("off")

plt.subplot(1,3,3)
plt.title("Low Resolution")
plt.imshow(low_up, cmap="gray")
plt.axis("off")

plt.show()
```

```
plt.show()
```



Task 4: Image Quantization (Gray Level Reduction)

```
import numpy as np

import matplotlib.pyplot as plt


# Quantization function

def quantize(img, levels):

    return np.floor(img / (256 / levels)) * (256 / levels)


# Quantized images

q256 = quantize(gray, 256)

q16 = quantize(gray, 16)

q4 = quantize(gray, 4)


# Display results

plt.figure(figsize=(12,6))


plt.subplot(1,3,1)

plt.title("256 Gray Levels")

plt.imshow(q256, cmap="gray")

plt.axis("off")
```

```
plt.subplot(1,3,2)
plt.title("16 Gray Levels")
plt.imshow(q16, cmap="gray")
plt.axis("off")

plt.subplot(1,3,3)
plt.title("4 Gray Levels")
plt.imshow(q4, cmap="gray")
plt.axis("off")

plt.show()
```

Task 5: Quality Observation & Analysis

```
import matplotlib.pyplot as plt

# Comparison figure
plt.figure(figsize=(14,8))

plt.subplot(2,3,1)
plt.title("High Resolution")
plt.imshow(high, cmap="gray")
plt.axis("off")

plt.subplot(2,3,2)
plt.title("Medium Resolution")
plt.imshow(medium_up, cmap="gray")
plt.axis("off")

plt.subplot(2,3,3)
plt.title("Low Resolution")
plt.imshow(low_up, cmap="gray")
plt.axis("off")

plt.subplot(2,3,4)
plt.title("16 Gray Levels")
plt.imshow(q16, cmap="gray")
plt.axis("off")

plt.subplot(2,3,5)
plt.title("4 Gray Levels")
```

```
plt.imshow(q4, cmap="gray")
plt.axis("off")

plt.show()



# Observations
print("\n--- Quality Observations ---\n")

print("1. Fine text details:")
print("   Lower resolution and heavy quantization remove thin strokes and edges.")

print("\n2. Readability:")
print("   Medium quality is readable, but low resolution and 4-level quantization")
print("   significantly reduce clarity.")

print("\n3. OCR suitability:")
print("   High resolution with full gray levels is best for OCR.")
print("   Degraded images increase recognition errors.")
```

High Resolution

Medium Resolution

Low Resolution

16 Gray Levels

4 Gray Levels

High Resolution

Medium Resolution

Low Resolution

16 Gray Levels

4 Gray Levels